

Install docker compose and create project structure

Note: We'll reuse app.py, requirements.txt from Practical 13

```
priyansh@IDEAPAD-18UT:~$ sudo apt install docker-compose-plugin -y
docker compose version
[sudo] password for priyansh:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker-compose-plugin is already the newest version (5.1.3-1~ubuntu.24.04~no
ble).
The following packages were automatically installed and are no longer requir
ed:
  linux-cloud-tools-6.8.0-106 linux-cloud-tools-6.8.0-106-generic
  linux-tools-6.8.0-106 linux-tools-6.8.0-106-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 86 not upgraded.
Docker Compose version v5.1.3
priyansh@IDEAPAD-18UT:~$ mkdir ~/compose-demo && cd ~/compose-demo
priyansh@IDEAPAD-18UT:~/compose-demo$
```

Use the given script in the corresponding file types

```
priyansh@IDEAPAD-18UT:~/compose-demo$ nano app.py
priyansh@IDEAPAD-18UT:~/compose-demo$ nano requirements.txt
priyansh@IDEAPAD-18UT:~/compose-demo$ nano docker-compose.yml
priyansh@IDEAPAD-18UT:~/compose-demo$
```

app.py - script

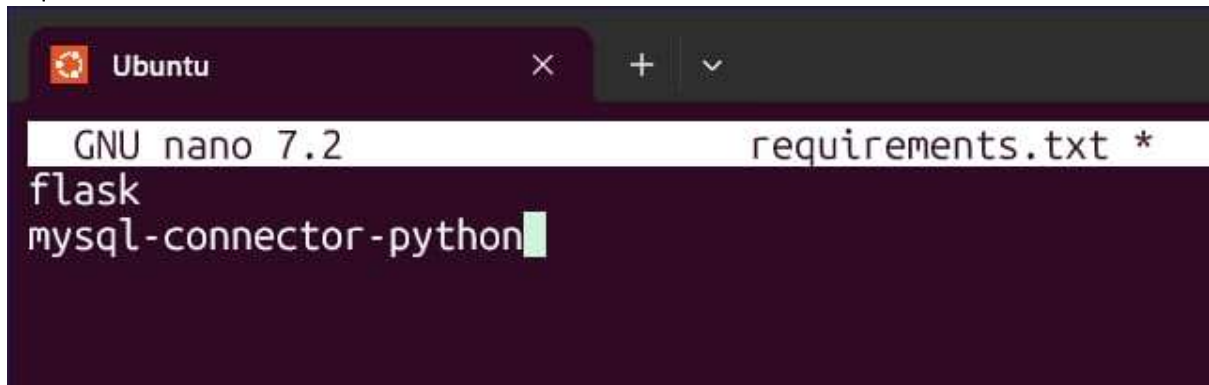
```
GNU nano 7.2 app.py *
from flask import Flask
import mysql.connector, os

app = Flask(__name__)

@app.route('/')
def home():
    try:
        conn = mysql.connector.connect(
            host=os.getenv('DB_HOST', 'db'),
            user=os.getenv('DB_USER', 'root'),
            password=os.getenv('DB_PASSWORD', 'rootpass'),
            database=os.getenv('DB_NAME', 'testdb')
        )
        return "<h1>Connected to MySQL!</h1>"
    except Exception as e:
        return f"<h1>App Running - DB error: {e}</h1>"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

requirements.txt

A screenshot of a terminal window with a dark background. The title bar shows 'Ubuntu' with a close button. The terminal header indicates 'GNU nano 7.2' and the file name 'requirements.txt *'. The file content is displayed in a light green monospace font, showing 'flask' on the first line and 'mysql-connector-python' on the second line, with a cursor at the end of the second line.

```
GNU nano 7.2 requirements.txt *
flask
mysql-connector-python
```

docker-compose.yml – script

A screenshot of a terminal window with a dark background. The title bar shows 'Ubuntu' with a close button. The terminal header indicates 'GNU nano 7.2' and the file name 'docker-compose.yml *'. The file content is displayed in a light green monospace font. It defines a 'web' service that builds from the current directory, listens on port 5000, and depends on a 'db' service. The 'db' service uses the 'mysql:8.0' image, sets environment variables for root password and database name, and maps a local volume to the database directory. A 'volumes' section at the bottom defines the 'db_data' volume.

```
GNU nano 7.2 docker-compose.yml *
version: '3.8'

services:
  web:
    build: .
    ports:
      - "5000:5000"
    environment:
      - DB_HOST=db
      - DB_USER=root
      - DB_PASSWORD=rootpass
      - DB_NAME=testdb
    depends_on:
      - db

  db:
    image: mysql:8.0
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: testdb
    volumes:
      - db_data:/var/lib/mysql
    ports:
      - "3306:3306"

volumes:
  db_data:
```

Start services, check them and logs for web service and DB

```
Ubuntu
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose up -d          # Start both services
docker compose ps           # Check running services
docker compose logs web     # Logs for web service
docker compose logs db      # Logs for DB
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 14/14
  ✓ Image mysql:8.0 Pulled 88.2s
[+] Building 0.7s (2/2) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 504B 0.0s
[+] up 14/15 [ ] load build definition from Dockerfile 0.1s
  ✓ Image mysql:8.0 Pulled 88.2s
  * Image compose-demo-web Building 1.2s
```

Scale up the instances

```
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose up -d --scale web=3
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 0.5s (2/2) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 504B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 2B 0.0s
[+] up 0/1
  * Image compose-demo-web Building 1.1s
```

Stop all, then stop and remove containers and volumes

```
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose stop
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose down
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose down -v
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
priyansh@IDEAPAD-18UT:~/compose-demo$
```

Once done, type exit to exit the terminal.